# SAHIL SAHRAN

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## Education

**University of Toronto** **Toronto, ON**
Bachelor of Applied Science in Electrical Engineering + PEY Co-op *Sep 2025 – Apr 2030 (expected)*
- cGPA: 3.97/4.00; Dean's List; ranked in the top 30 of the ECE cohort
- Relevant Coursework: Introductory Electronics (diodes, transistors, op-amps), Digital Systems (Verilog, FPGA design), Programming Fundamentals (C++)

## Experience

**Embedded Systems Intern** **Jun 2026 – Aug 2026**
Nanotronic Electronic *North York, ON*
- Drew Altium schematics for an internal RC aircraft transmitter (ESP32, RYLR993 LoRa module, USB-to-SPI joystick bridge), building every symbol and footprint from scratch with JLCPCB part numbers for direct BOM ordering
- Wired the load cell and wrote the firmware for a stepper-driven dispenser that dispenses by weight, testing NAU7802 and HX711 24-bit ADCs. Reduced noise from $\pm 1$ g to $\pm 50$ mg by proposing an LDO supply, which halved the noise on its own, then moving the decoupling caps to the ADC pins and enabling the PGA filter cap on the unused channel
- Assembled 100+ boards for a medical project using stencil paste reflow and hand soldering, then tested each one and fixed any shorted rails, cold joints, or reversed diodes

**CNC Programmer** **May 2026 – Sep 2026**
D2Woodworking *Markham, ON*
- Programmed and ran CNC cuts in Mozaik CAD/CAM from architectural drawings for 3 whole-house and 5+ smaller projects (over $100K CAD), often working alone in the shop, where any cutting error meant wasted material, rework, and missed install deadlines
- Built parametric cabinet templates driven by algebraic constraints that adjust in all three dimensions to customer specifications, eliminating manual recalculation on every order

## Projects

**ESPRA: ESP32 Speech and Proximity Roaming Assistant | ESP32-S3, C++, FreeRTOS, Python** **Jun 2026 – Oct 2026**
- Traced persistent audio buzz to the speaker wiring and rewired the two 4 Ω speakers from parallel (2 Ω, below the MAX98357A's rated load) to series (8 Ω), then capped digital audio at 0.25–0.40 of full scale to stop clipping from the amp's fixed +9 dB gain
- Split audio and graphics across the ESP32-S3's two cores with FreeRTOS, so blocking SSD1309 OLED writes over 400 kHz I2C no longer interrupted the 24 kHz APLL-clocked I2S stream. Zero-filling the DMA buffer stopped pops during redraws, and rate-limiting the mouth animation removed I2C buzz from the speaker leads
- Added a VL53L0X ToF sensor that wakes it when someone sits down and dual INMP441 mics for TDOA direction finding, streaming speech to a Python server (Groq Whisper, LLM, Edge-TTS) over 4 KB framed USB serial and later HTTPS, replying in under 5 s

**Beehive Winter Monitoring System | Client: UofT BEES, team of 6** **Jan 2026 – Apr 2026**
- Led the electrical design of a ∼$200 winter hive monitor, placing SCD41 $CO_2$ and BME680 sensors in a vented in-hive enclosure and three DS18B20 probes on a shared 1-Wire bus, cabled through IP68 glands to an outdoor ESP32 enclosure uploading data over Wi-Fi every hour
- Derived a 1.94 V/m E-field exposure limit from a published bee EMF stress study, then used free-space power density and the 377 Ω impedance of free space to calculate an 11.2 cm minimum ESP32 distance at 2 dBm, confirmed by EMF meter readings of ∼1 V/m

**Size-Targeted JPEG Compression | Independent Research, MATLAB** **Aug 2024 – Nov 2024**
- Modified JPEG so the user sets the output file size instead of a quality level, using Lagrange multipliers to choose the quantization table scale and zero-rounding threshold that minimize mean squared error at that size
- Developed an original file size estimation method by modeling DCT coefficients with a Laplacian distribution and deriving the average Huffman code length from the ITU-T T.81 tables, predicting compressed size within 5.6% on average across 5 test images

## Skills

**Programming:** C, C++, Python (NumPy), MATLAB, Verilog
**Hardware:** Altium Designer, LTspice, Quartus Prime, soldering (through-hole, SMD stencil/reflow), board-level debugging
**Embedded:** ESP32 (Arduino framework), FreeRTOS, I2C, I2S, SPI, UART, 1-Wire, DMA audio streaming, Wi-Fi web servers
**CAD & Languages:** Autodesk Fusion, SketchUp, Mozaik CAD/CAM | English (native), French (working proficiency)